

PROJECT REPORT

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Built on the ServiceNow Now Platform

Team ID	SWTID-2026-9879
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Platform	ServiceNow (Service Catalog, Flow Designer)
Date	29 September 2026

Team Members

S.No	Team Member	Role in the Project
1	Pradeepa K R	Catalog item design, Catalog UI Policy, My Requests view
2	Priyadharshan M C	Request submission, budget limit property, auto-approval flow, task closure, reports
3	Priyadharshini V	Department Head approval flow, catalog task creation, dashboard
4	Priyadharshini V	Documentation, end-to-end testing support
5	Ragul P	Inventory table, stock check, stock update, notifications, testing

Table of Contents

1. Introduction

- 1.1 Project Overview
- 1.2 Purpose

2. Ideation Phase

- 2.1 Problem Statement
- 2.2 Empathy Map Canvas
- 2.3 Brainstorming

3. Requirement Analysis

- 3.1 Customer Journey Map
- 3.2 Solution Requirement
- 3.3 Data Flow Diagram
- 3.4 Technology Stack

4. Project Design

- 4.1 Problem Solution Fit
- 4.2 Proposed Solution
- 4.3 Solution Architecture

5. Project Planning & Scheduling

- 5.1 Project Planning

6. Functional and Performance Testing

- 6.1 Performance Testing

7. Results

- 7.1 Output Screenshots

8. Advantages & Disadvantages

9. Conclusion

10. Future Scope

11. Appendix

1. INTRODUCTION

1.1 Project Overview

In most organizations, an employee who needs a standard laptop raises a request that is approved and processed manually by the IT team. Approvers are interrupted even for routine, within-budget orders, IT staff spend time creating tasks and checking stock by hand, and requesters get no clear update on where their order stands. The result is slow delivery, inconsistent handling and a steady stream of follow-up queries.

This project builds a Standard Laptop Order catalog item in the ServiceNow Service Catalog and automates its entire life cycle with Flow Designer. The employee fills in the laptop type (Developer, Business or Basic), brand, screen size and accessories; a Catalog UI Policy shows only the RAM and storage options that match the chosen type. A Flow Designer flow then compares the order cost with the approved budget limit. Orders within the limit are auto-approved, while orders above it are sent to the Department Head. After approval, the flow checks stock in the Inventory table, creates a catalog task (SCTASK) for the IT Support group and sends stage-wise email notifications (placed, approved or rejected, delivered) to the requester. A dashboard reports pending orders, approval rate and average delivery time.

The complete solution is configured with native ServiceNow low-code features, with no custom code or external tools, so the budget limit, approvers and laptop options can be changed later without rebuilding the flow.

1.2 Purpose

The purpose of this project is to streamline IT procurement by replacing the manual handling of standard laptop requests with an automated, transparent workflow. Specifically, the project aims to:

- Reduce processing time and manual effort by automating repetitive approval, task-creation and stock-checking steps.
- Improve accuracy and consistency by making sure every standard request follows the required configuration and approval rules.
- Increase visibility by giving the requester a request number, a status view under My Requests and email updates at every stage.
- Free Department Heads from routine approvals so that they review only the orders that exceed the budget or need a special decision.
- Reduce the repetitive workload of the IT Support team so that it can focus on more critical issues.
- Create a scalable procurement workflow that can later be extended to other standard requests.

2. IDEATION PHASE

2.1 Problem Statement

Today, employees who need a standard laptop submit a request that is approved and processed manually by the IT team. Approvers are interrupted even for routine, within-budget orders, IT staff spend time creating tasks and checking stock, and requesters receive no clear updates on their order. This leads to slow delivery, inconsistent handling and frequent follow-up queries. We need a Standard Laptop catalog item in ServiceNow, powered by Flow Designer, that auto-approves orders within the budget limit, sends the rest to the Department Head, automatically creates the fulfillment task for the IT Support team, and keeps the requester informed with a request number and stage-wise notifications.

Customer Problem Statements

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A new joiner or employee who needs a standard laptop to start work.	Place a laptop request in the Service Catalog and receive the device without delay.	The request sits in a queue until someone approves it, and I have no idea where it is stuck.	Approvals and updates are handled manually, and no status notifications are sent to me.	Anxious and unable to plan my work because the delivery date is unknown.
PS-2	An IT support / fulfillment team member.	Fulfill approved laptop orders correctly and within the target time.	I have to read each request, create the task, check stock and update the asset record by hand.	Every routine order needs the same repetitive steps, so mistakes and missed requests creep in.	Stressed and stretched thin, with little time left for critical issues.
PS-3	A Department Head who approves equipment purchases.	Review only the requests that exceed the budget or need a special decision.	My inbox fills with approval requests for standard laptops that are within the limit.	No rule exists to auto-approve standard, within-budget orders.	Distracted and annoyed that my time is spent on routine approvals.

2.2 Empathy Map Canvas

The empathy map captures what Meera, a new joiner who needs a standard laptop, says, thinks, does and feels while waiting for her order under the current manual process. It guided the team in designing a Flow Designer based ordering flow with instant approval for in-budget requests, automatic task creation for IT Support and stage-wise status updates to the requester.

MEERA (New joiner requesting a standard laptop)	
SAYS "My joining date was Monday. Has my laptop request been approved yet?" "Why is there no way to track my order?" "I can't start my training modules without a laptop."	THINKS "Is my request stuck with a manager who is on leave?" "A standard laptop shouldn't need this much waiting." "I wish I got a message at every step."
DOES - Submits the laptop request in the Service Catalog on day one - Messages the IT help desk and her team lead to ask for updates - Borrows a colleague's system or works from a personal device meanwhile	FEELS - Impatient and left out while others are already working - Worried about falling behind in her first weeks - Reassured when she gets a clear confirmation and status update

2.3 Brainstorming

Step 1: Brainstorm and idea listing. Each team member listed ideas for improving how employees order laptops, without judging them at this stage.

S.No	Team Member	Idea / Suggestion	Category	Group
1	Pradeepa K R	Create a Laptop Order catalog item in the Service Catalog with variables such as Laptop Type (Developer / Business / Basic), Brand, Screen Size and Accessories. Use a Catalog UI Policy to show only the RAM and storage options that match the selected Laptop Type.	Catalog Design	1
2	Priyadharshan M C	Use a Flow Designer 'If' condition to auto-approve the order when the request cost is within the approved limit.	Approval Automation	2
3	Priyadharshini V	Send an approval request to the Department Head when the cost is above the limit, and update the requester with the rejection comment if the request is rejected.	Approval Automation	2
4	Priyadharshini V	Once approved, generate a Catalog Task and assign it to the IT Support team; use the Inventory table to check stock availability.	Fulfillment Automation	2
5	Ragul P	Send automatic email and in-platform notifications at the placed, approved, rejected and delivered stages, and build a dashboard showing pending orders, approval rate and average delivery time.	Notifications	3

Step 2: Idea prioritization. The grouped ideas were rated on feasibility and importance, and the final ideas were selected to move forward.

Group	Final Idea	Feasibility	Importance	Priority	Selected
1	Standard Laptop catalog item with request variables	High	High	1	Yes
2	Flow Designer flow for approval, task creation and fulfillment	High	High	1	Yes
3	Requester notifications and order status reporting	Medium	Medium	2	Yes

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey map follows an employee who orders a laptop through the Laptop Order catalog item, from submitting the request to collecting the laptop from the IT Support team. For each stage it records actions, touchpoints, thoughts and feelings, the team members who own the step, and opportunities for improvement through Flow Designer automation.

Phase	Stage 1: Request	Stage 2: Approval	Stage 3: Fulfillment
Actions	- Logs in to ServiceNow and opens the Service Catalog - Opens Laptop Order and chooses laptop type, brand and screen size - Adds accessories and submits the request	- Receives the request number - Waits while the request is auto-approved or sent to the Department Head - Checks the status under My Requests	- Gets notified that the laptop is ready - Collects the laptop from the IT Support team - Verifies the configuration and starts working
Touchpoints	- Service Catalog home page - Laptop Order catalog item form - Submit button	- Request (REQ) record - Approval record for the Department Head - My Requests status page	- Catalog Task (SCTASK) assigned to IT Support - Inventory table and asset record - Delivery notification email
Customer Thoughts	"Which laptop type suits my role?" "Will the RAM and storage match my needs?" "I hope I get a request number."	"Has my request been approved?" "Who is reviewing it?" "When will I get an update?"	"When can I collect my laptop?" "Is it set up as I asked?" "That was simple and quick."
Customer Feelings	Interested but a little unsure; confident once only matching RAM and storage options appear.	Anxious while waiting; reassured by the request number and status updates.	Happy and relieved when the laptop is delivered on time.
Process Ownership	Catalog item and UI Policy: Pradeepa K R Request submission and budget limit: Priyadharshan M C	Auto-approval flow: Priyadharshan M C Department Head approval flow: Priyadharshini V	Catalog task creation: Priyadharshini V Inventory check and notifications: Ragul P
Opportunities	- Add laptop type presets for different roles - Show approximate cost and specs on the form - Make justification mandatory for non-standard choices	- Auto-approve orders within the budget limit - Ask the Department Head only above the limit - Show live status to the requester	- Send an email when the laptop is ready - Auto-close the request after the task closes - Report on average delivery time

3.2 Solution Requirement

Functional requirements describe what the solution does (accepting an order, routing it for approval, raising a catalog task). Non-functional requirements describe how well it does so, in terms of usability, security, speed and maintainability.

Functional Requirements (FR)

S.No	Category	Requirement Description	Priority
1	Authentication	Every user logs in to the ServiceNow instance with personal credentials before opening the Service Catalog. The system records the person as the requester through the Requested for / Opened by fields.	High
2	Authorization levels	Requester: any employee with catalog access can place a laptop order. Department Head: approves or rejects orders above the budget limit. IT Support group: works on the fulfillment catalog tasks. Administrator: builds and maintains the catalog item and the flow.	High
3	External interfaces	No third-party system or external API is used; the solution runs on native Service Catalog and Flow Designer features.	Low
4	Transactions processing	Placing an order generates a Request (REQ) with a Requested Item (RITM). The flow starts from the Service Catalog trigger, checks stock availability and runs the Create Catalog Task action to raise an SCTASK for IT Support.	High
5	Reporting	The requester follows progress under My Requests and sees the request number. A dashboard shows pending orders, approval rate and average delivery time; flow run history is available in Flow Designer.	Medium
6	Business rules	Laptop Order offers laptop type (Developer, Business, Basic), brand and screen size, with RAM and storage filtered by a Catalog UI Policy, plus optional accessories. Orders within the budget limit are auto-approved; orders above it go to the Department Head. After approval a catalog task is assigned to IT Support.	High
7	Compliance	Employee and request data stays inside the organization's own ServiceNow instance and is guarded by roles and access controls.	Low
8	Other	The catalog item is linked to the flow through its Process Engine tab, and the flow must be activated before any order can trigger it.	High

Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance
1	Performance & Speed	After an order is submitted and approved, the flow raises the fulfillment task on its own, with no manual step.	Task appears within 2 minutes after approval
2	Scalability	Every order starts its own flow run, so several requests are handled in parallel.	Parallel orders each get their own task
3	Security & Data Privacy	Only logged-in users can order; request and task records are visible only to users with appropriate roles.	Mandatory login; role-based access on every record
4	Reliability & Availability	The flow runs inside the ServiceNow platform and its execution history can be checked when an error occurs.	Each order results in a request, an item and a task
5	Usability & Accessibility	Clear field labels, simple selection lists and a confirmation message showing the request number.	Order submitted in a few clicks; confirmation displayed
6	Maintainability	The flow is built visually in Flow Designer without custom code, so administrators can change it easily.	No custom scripting

3.3 Data Flow Diagram

The diagram traces how data moves in the laptop ordering system. The employee places an order through the catalog, the Flow Designer approval logic evaluates the cost, and the IT Support Team delivers the laptop once stock is confirmed.

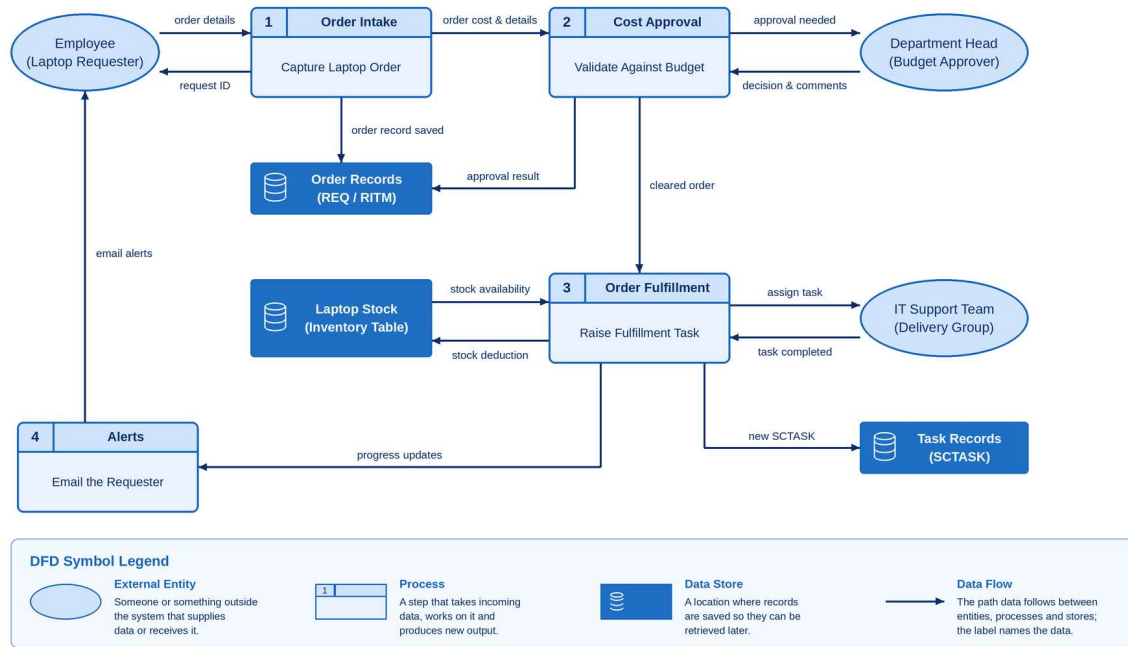


Figure 3.1: Data Flow Diagram - Laptop Order Approval and Fulfillment

No.	Process	Description
1	Order Intake	The employee opens Laptop Order in the Service Catalog, picks the laptop type, brand and screen size, and submits the form. The system generates a Request (REQ) and a Requested Item (RITM) and returns the request number.
2	Cost Approval	The flow compares the order cost with the approved budget. Within the limit, the order is cleared automatically. Above the limit, it goes to the Department Head, who approves or rejects it with comments.
3	Order Fulfillment	Once cleared, the flow looks up the Laptop Stock table and raises a catalog task (SCTASK) for the IT Support Team. On delivery the task is closed, the stock count is reduced and the request is marked complete.
4	Alerts	Whenever the order is placed, approved or rejected, and delivered, the requester is sent an email, so there is no need to chase for updates.

3.4 Technology Stack

The architecture below shows how a standard laptop order moves through the ServiceNow platform, from the employee's request to approval, fulfillment and notification, all inside ServiceNow.

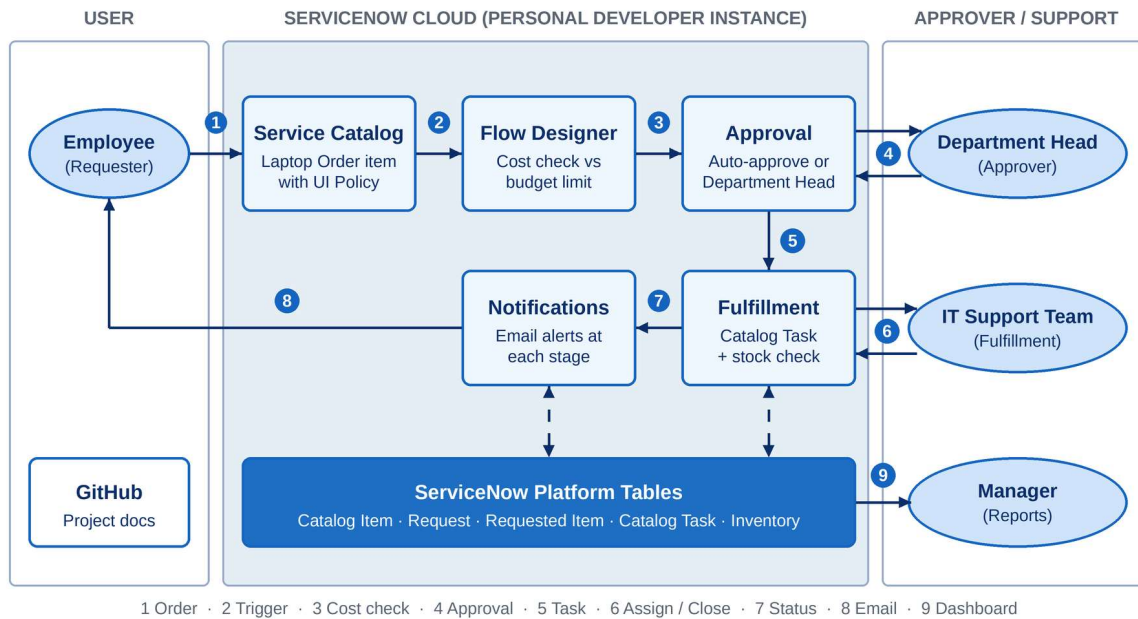


Figure 3.2: Technical architecture - Automated Standard Laptop Ordering on ServiceNow

Table 1: Components and Technologies

S.No	Component	Description	Technology
1	User Interface	Employees place laptop orders through the Service Catalog form with dynamic RAM and storage options.	Service Catalog, Catalog Item variables, Catalog UI Policy
2	Application Logic-1	Compares the order cost with the budget limit and routes the request: auto-approve or send to the Department Head.	Flow Designer (If condition, Ask for Approval)
3	Application Logic-2	After approval, creates the catalog task for IT Support and checks stock in the Inventory table.	Create Catalog Task action, Inventory table lookup
4	Application Logic-3	Sends stage-wise email notifications (placed, approved / rejected, delivered) to the requester.	ServiceNow Notifications (email)
5	Database	Stores the catalog item, request, requested item, catalog task and laptop stock records.	Platform tables (Catalog Item, REQ, RITM, SCTASK, Inventory)
6	Cloud Database	Tables are hosted in the ServiceNow cloud instance; no separate database service is needed.	Now Platform database
7	File Storage	Project documents and screenshots are kept in a shared repository; request attachments use platform storage.	GitHub; ServiceNow Attachments
8	External API	No external API is required.	None
9	Machine Learning Model	Not used in this project.	Not applicable
10	Infrastructure	Cloud-hosted ServiceNow instance; no local server configuration required.	ServiceNow Personal Developer Instance (PDI)

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	No open-source framework; built with ServiceNow's native low-code features.	Now Platform (low-code)
2	Security Implementations	Mandatory login; access to requests and tasks is limited by roles; data stays in the organization's instance.	RBAC, ACLs, instance authentication
3	Scalable Architecture	Every order runs in its own flow execution, so many requests are handled in parallel.	Flow Designer on Now Platform cloud
4	Availability	The instance is hosted and maintained by ServiceNow; flow execution history helps diagnose failures.	ServiceNow cloud hosting, execution history
5	Performance	Flows run server-side and the task is created within about 2 minutes of approval; dashboards give a quick view of order status.	Server-side flow execution, Reports and Dashboards

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Problem-Solution Fit canvas shows how the Standard Laptop ordering flow fits the real problem of employees who wait for manually handled laptop requests. It lists the customer segment, problems and triggers, available alternatives, our solution, and the channels and behavior that support it.

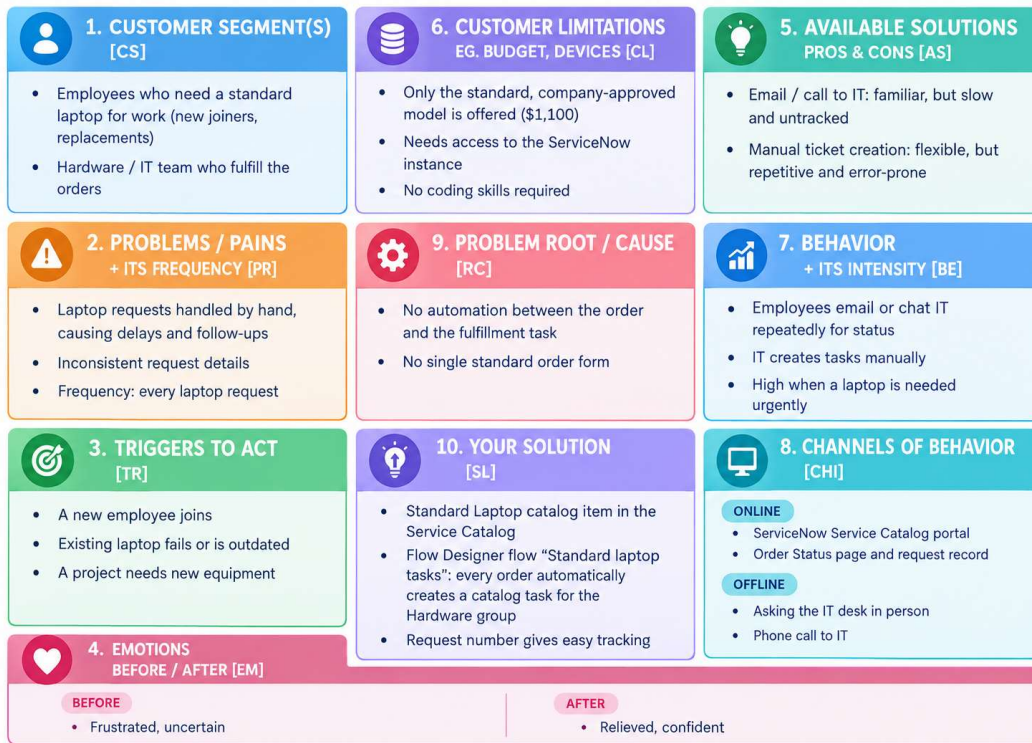


Figure 4.1: Problem-Solution Fit canvas

Purpose of the fit

- Streamline IT procurement by automating standard laptop orders using Flow Designer.
- Reduce processing time and manual effort by automating repetitive procurement tasks and workflows.
- Improve accuracy and consistency by making standard requests follow the required configuration and approval process.
- Increase visibility and transparency with clear request status and fewer unnecessary follow-ups.
- Minimize delays and errors caused by manual checking, task creation and request handling.
- Automate approval and task routing so requests reach the right team without manual intervention.

4.2 Proposed Solution

S.No	Parameter	Description
1	Problem Statement	Employees who need a laptop raise a request that is approved and processed manually. Approvers are interrupted for routine in-budget orders, IT staff create tasks and check stock by hand, and requesters get no clear status updates, causing slow delivery and repeated follow-ups.

S.No	Parameter	Description
2	Idea / Solution Description	A Laptop Order catalog item with variables for laptop type, brand, screen size and accessories. A Catalog UI Policy shows only matching RAM and storage options. A Flow Designer flow compares the order cost with the approved budget: in-budget orders are auto-approved, higher-cost orders go to the Department Head. After approval the flow checks stock in the Inventory table, creates a catalog task for the IT Support group and sends email notifications at each stage. A dashboard shows pending orders, approval rate and average delivery time.
3	Novelty / Uniqueness	Combines cost-based auto-approval, stock-aware task creation and stage-wise notifications in one low-code flow, so a standard order needs no manual step from approvers or IT staff. Built with native ServiceNow features (no custom code), so the budget limit, approvers and options are easy to change.
4	Social Impact / Customer Satisfaction	Employees get laptops faster and can see order status, which reduces waiting and anxiety, especially for new joiners. Managers are no longer interrupted by routine approvals, and IT staff can focus on complex support issues.
5	Business Model	An internal IT service improvement with no direct revenue. Value comes from saved IT and approver time, fewer onboarding delays and better control of laptop spending and stock. The catalog item and flow can be packaged as a reusable update set and offered to other departments or client organizations as an implementation or managed service.
6	Scalability	Every order runs in its own flow execution, so many requests are handled in parallel. Budget limit, laptop types and approvers can be changed without rebuilding the flow, and the pattern extends to monitors, mobiles, software licenses and more departments or locations.

4.3 Solution Architecture

Solution architecture bridges the gap between the business problem and the technology solution. Here it describes how the Service Catalog, Flow Designer, notifications, fulfillment and platform tables work together to process a laptop order, and defines the two approval paths.

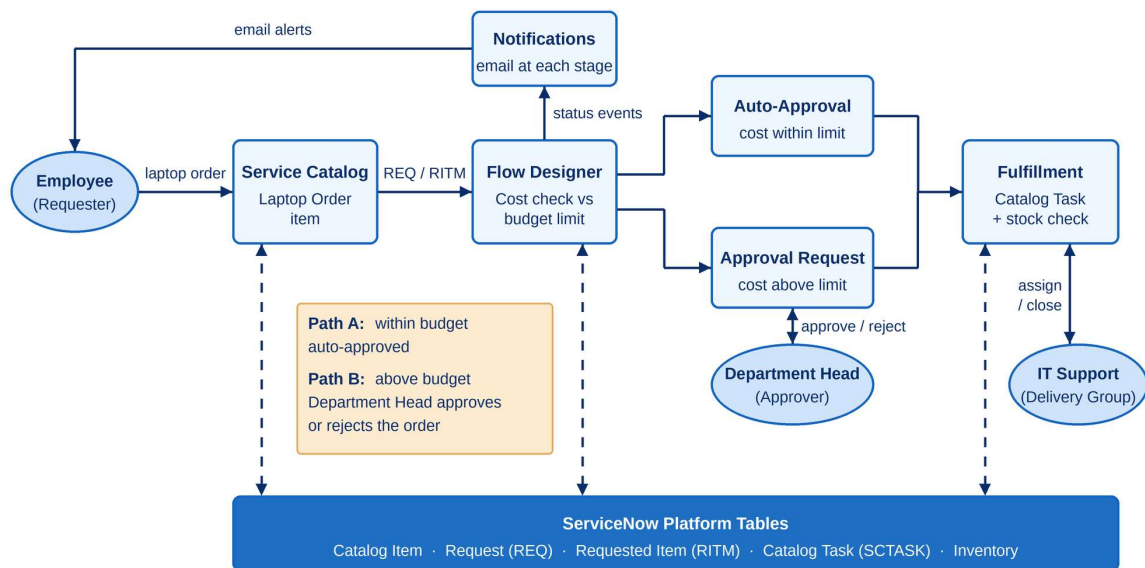


Figure 4.2: Architecture and data flow of the automated standard laptop ordering solution

Path A: within budget, the order is auto-approved and goes straight to fulfillment. Path B: above budget, the Department Head approves or rejects the order; approved orders proceed to fulfillment, where a catalog task is assigned to IT Support after the stock check.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work is organized into four sprints of six days each, with 20 story points per sprint (80 in total). Each sprint covers one functional epic.

Product Backlog, Sprint Schedule and Estimation

Sprint	Epic	USN	User Story / Task	Pts	Priority	Team Member
Sprint-1	Catalog Design	USN-1	As an employee, I can open the Standard Laptop catalog item and select the laptop type, brand and screen size.	3	High	Pradeepa K R
Sprint-1		USN-2	As an employee, I see only the RAM and storage options that match my selected laptop type (Catalog UI Policy).	5	High	Pradeepa K R
Sprint-1		USN-3	As an employee, I can add accessories and a business justification to my laptop request.	3	Medium	Pradeepa K R
Sprint-1		USN-4	As an employee, I can submit the request and receive a request number for tracking.	2	High	Priyadharshan M C
Sprint-1	Configuration & Inventory	USN-5	As an admin, I can set the approved budget limit as a system property used by the flow.	2	Medium	Priyadharshan M C
Sprint-1		USN-6	As an IT support member, I can maintain an Inventory table with laptop models and stock count.	5	High	Ragul P
Sprint-2	Approval Automation	USN-7	As an employee, my request is automatically approved when the cost is within the approved limit (Flow Designer).	5	High	Priyadharshan M C
Sprint-2		USN-8	As a Department Head, I receive an approval request only when the cost is above the limit.	5	High	Priyadharshini V
Sprint-2		USN-9	As a Department Head, I can approve or reject a request and add comments.	3	High	Priyadharshini V
Sprint-2		USN-10	As an employee, I am updated with the rejection comment and request status if my request is rejected.	2	Medium	Priyadharshini V
Sprint-2		USN-11	As a team, we test both the auto-approval and the Department Head approval paths.	5	Medium	Ragul P
Sprint-3	Fulfillment Automation	USN-12	As an IT Support member, a Catalog Task is created and assigned to my group once the order is approved.	5	High	Priyadharshini V
Sprint-3		USN-13	As an IT Support member, I can check stock availability in the Inventory table before fulfilling the order.	5	High	Ragul P
Sprint-3		USN-14	As an IT Support member, the stock count and asset record are updated when the laptop is delivered.	3	Medium	Ragul P
Sprint-3		USN-15	As an IT Support member, I can close the task so that the request is marked Closed Complete.	2	High	Priyadharshan M C
Sprint-3		USN-16	As an employee, I can view the current stage of my order under My Requests.	2	Medium	Pradeepa K R
Sprint-3		USN-17	As an IT Support member, an out-of-stock order is flagged as pending procurement.	3	Low	Priyadharshan M C
Sprint-4	Notifications & Reporting	USN-18	As an employee, I receive an email when my laptop request is placed.	3	High	Ragul P
Sprint-4		USN-19	As an employee, I receive an email when my request is approved or rejected.	3	High	Ragul P
Sprint-4		USN-20	As an employee, I receive an email when my laptop is delivered.	2	Medium	Pradeepa K R

Sprint	Epic	USN	User Story / Task	Pts	Priority	Team Member
Sprint-4		USN-21	As a manager, I can view a dashboard showing pending orders and approval rate.	5	Medium	Priyadharshini V
Sprint-4		USN-22	As a manager, I can view a report of the average delivery time for laptop orders.	3	Medium	Priyadharshan M C
Sprint-4		USN-23	As a team, we complete end-to-end testing and prepare the project documentation.	4	High	All team members

Project Tracker, Velocity and Burndown Chart

Sprint	Total Story Points	Duration	Start Date	Planned End Date	Points Completed (at planned end)	Actual Release Date
Sprint-1	20	6 Days	05 Oct 2026	10 Oct 2026		
Sprint-2	20	6 Days	12 Oct 2026	17 Oct 2026		
Sprint-3	20	6 Days	19 Oct 2026	24 Oct 2026		
Sprint-4	20	6 Days	26 Oct 2026	31 Oct 2026		

Velocity

With a sprint duration of 6 days and a team velocity of 20 story points per sprint, the average velocity (AV) per day is $AV = \text{Sprint Velocity} / \text{Sprint Duration} = 20 / 6 \approx 3.33$ story points per day. Over the four sprints the team plans to deliver 80 story points in 24 working days.

Burndown Chart

The burndown chart plots the remaining story points against the sprint days. The line below shows the planned (ideal) burn of 80 points; the actual line can be added as the sprints are completed.

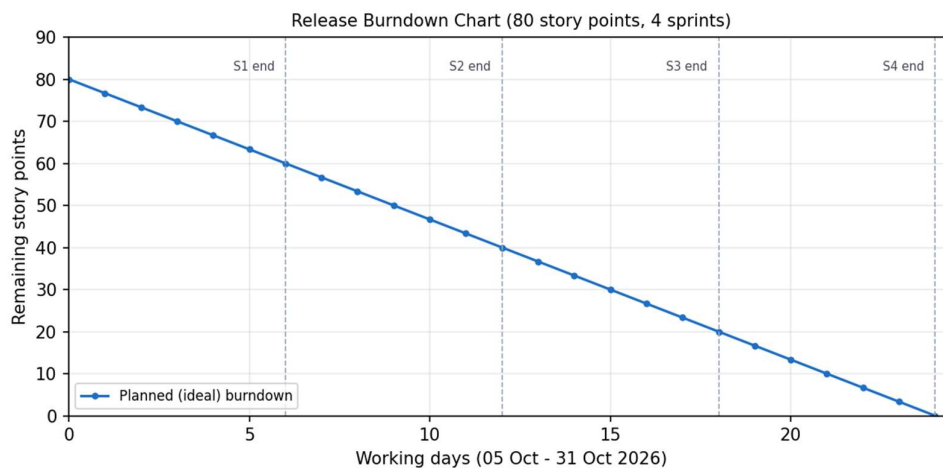


Figure 5.1: Planned burndown chart for the four sprints

6. FUNCTIONAL AND PERFORMANCE TESTING

Functional testing verifies that every step of the laptop ordering flow behaves as specified. The test cases below cover the catalog form, both approval paths, fulfillment, notifications and reporting. The Actual Result and Status columns are filled in during the testing in Sprint-4 (USN-23).

Functional test cases

TC ID	Test Scenario	Steps	Expected Result	Actual	Status
TC-01	Open Laptop Order item	Log in as employee, open Service Catalog, open Laptop Order.	Form shows laptop type, brand, screen size, accessories and justification.		
TC-02	Catalog UI Policy	Select laptop type Developer, then Basic.	Only the RAM and storage options for the selected type are shown.		
TC-03	Submit request	Fill the form and click Submit.	REQ and RITM are created; request number is displayed.		
TC-04	Auto-approval	Order a laptop with cost within the budget limit.	Order is approved automatically; no approval sent to the Department Head.		
TC-05	Approval routing	Order a laptop with cost above the limit.	Approval request is generated for the Department Head.		
TC-06	Rejection	Department Head rejects with a comment.	Request is rejected; requester receives the comment and status.		
TC-07	Approval	Department Head approves the request.	Flow continues to stock check and task creation.		
TC-08	Task creation	Order approved with stock available.	SCTASK is created and assigned to the IT Support group.		
TC-09	Out of stock	Order a model whose stock count is 0.	Order is flagged as pending procurement.		
TC-10	Delivery and closure	IT Support closes the task.	Request is Closed Complete; stock count is reduced by 1.		
TC-11	Notifications	Run an order through placed, approved/rejected and delivered stages.	Requester receives an email at each stage.		
TC-12	Dashboard and report	Open the dashboard and average delivery time report.	Pending orders, approval rate and average delivery time are displayed.		
TC-13	Budget limit property	Change the budget limit system property and place two orders.	Routing follows the new limit without editing the flow.		

6.1 Performance Testing

Performance testing checks the non-functional requirements defined in Section 3.2 against their target metrics.

S.No	Parameter	Test Method	Target Metric	Observed	Status
1	Task creation time	Approve an order and note the time at which the SCTASK appears.	Within 2 minutes of approval		
2	Parallel orders	Submit 5 orders at the same time from different users.	Each order gets its own flow run and task		
3	Form usability	Count clicks needed to place an order.	Order placed in a few clicks; confirmation shown		

S.No	Parameter	Test Method	Target Metric	Observed	Status
4	Reliability	Check Flow Designer execution history for all test orders.	No failed execution; REQ, RITM and SCTASK exist for each order		
5	Security	Open request and task records as a user without the required role.	Access denied; mandatory login enforced		
6	Notification delivery	Check the email log after each stage.	Email sent at every stage		

7. RESULTS

7.1 Output Screenshots

The screenshot shows the 'Standard laptop task' flow configuration in Workflow Studio. The flow is currently 'Inactive'. The configuration includes:

- Table Name:** Catalog Task [sc_task]
- Trigger:** Trigger -> Requested Item ...
- Short Description:** Laptop need to Configured
- Description:** Laptop need to configured
- Assignment group:** Hardware
- Approval:** Approved
- Wait:** Checked
- Template Catalog Item:** Select Template Catalog Item
- Available:** Select Template Catalog Item field to load variables.
- Selected:** Create Catalog Item(s) that point to this flow and the variables on that item(s) will show here.

The right sidebar shows the 'Data' section with 'Flow Variables' and 'Trigger - Service Catalog' variables. The '1 - Create Catalog Task' section shows 'Catalog Task' and 'Action Status' variables.

Buttons at the bottom: Delete, Cancel, Done.

Figure 7.1: Create Catalog Task – Flow Action Configuration

The screenshot shows the 'Standard laptop task' flow configuration in Workflow Studio. A confirmation dialog is displayed over the configuration:

Are you sure you want to activate the flow?

Your flow will start running when the trigger conditions are met.

☐ Don't show me this again

Buttons: Cancel, Activate

The background shows the flow configuration with the 'TRIGGER' set to 'Service Catalog' and the 'ACTIONS' list containing '1 - Create Catalog Task'.

Figure 7.2: Flow Activation Confirmation

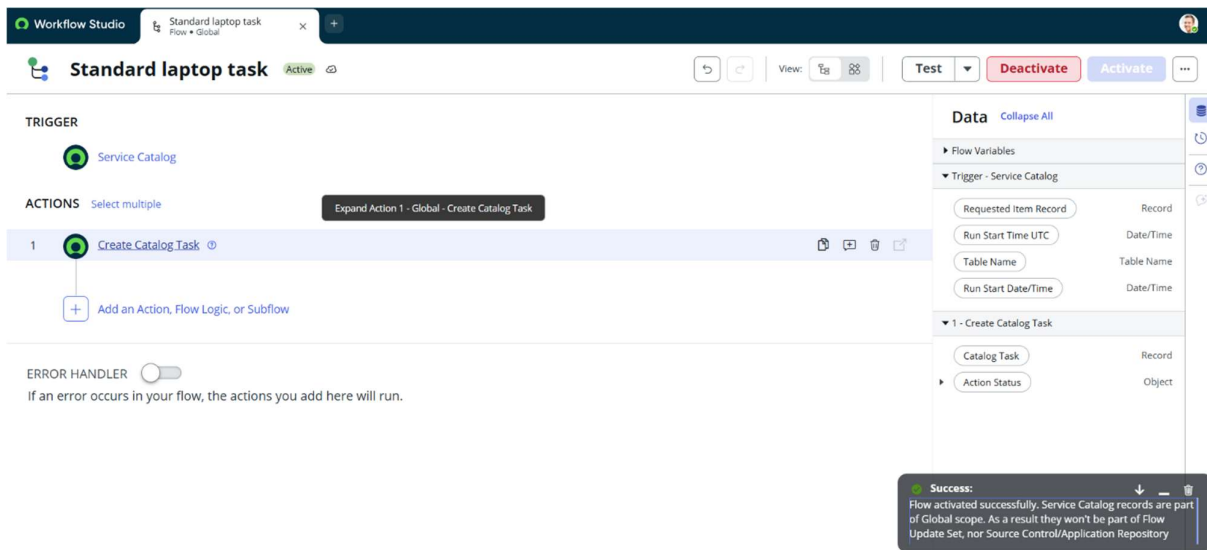


Figure 7.3: Flow Designer – Standard Laptop Task Automation

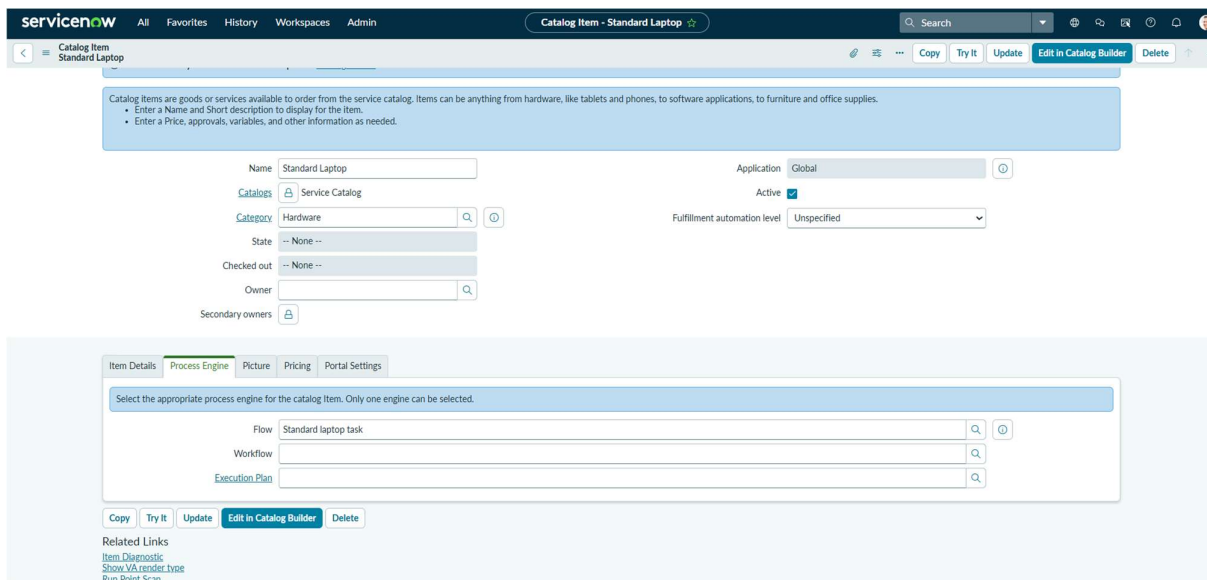


Figure 7.4: Standard Laptop Catalog Item Configuration

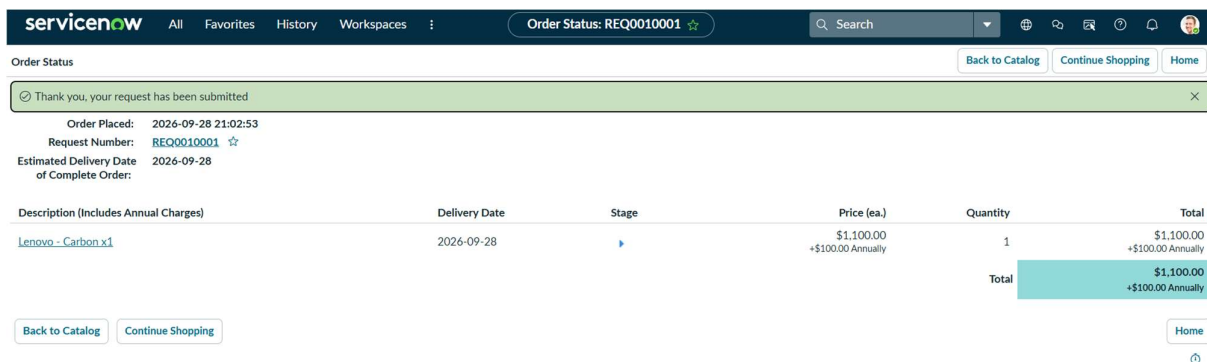


Figure 7.5: Standard Laptop Order Status Confirmation

servicenow All Favorites History Workspaces Admin Request - REQ0010001

Request: REQ0010001

Due date: 2026-09-28 21:02:52

Price: \$1,100.00

Description:

Short description:

Special instructions:

Comments (Customer visible):

Activities: 1

System Administrator

Impact: 3 - Low

Opened by: System Administrator

Priority: 4 - Low

State: Open

Field changes: 2026-09-28 21:02:53

Update Add New Item Cancel Request Copy Delete

Requested Items (1) Approvals (1) Group approvals (1) Recurring Prices (1)

Approval for - REQ0010001

State	Approver	Comments	Created
Approved	Eric Schroeder		2026-09-28 21:02:57

Figure 7.6: Request Details and Approval Status

Catalog Task SCTASK0010001

Number: SCTASK0010001

Assigned to:

Configuration Item:

Active: ☒

Approval: Approved

Priority: -- None --

State: Open

Request Item: RITM0010001

Requested for: System Administrator

Short description: Laptop need to Configured

Description: Laptop need to configured

Comments (Customer visible):

Activities: 1

System

Impact: 3 - Low

Opened by: [Empty]

State: Open

Field changes: 2026-09-28 21:03:40

Update Close Task Delete

Affected CIs Approvals Group approvals

Configuration Item Search

Task: SCTASK0010001

Configuration Item	Class	Support group	Owned by	Applied	Applied date	Manual proposed change	Updated
--------------------	-------	---------------	----------	---------	--------------	------------------------	---------

Figure 7.7: Catalog Task – Standard Laptop Configuration Request

8. ADVANTAGES & DISADVANTAGES

Advantages

- Faster delivery: in-budget orders are approved instantly and the fulfillment task is created automatically.
- Less manual effort: Department Heads see only above-limit orders, and IT staff no longer create tasks by hand.
- Transparency: the requester gets a request number, a status view under My Requests and stage-wise emails.
- Consistency: every order follows the same rules for configuration, approval and stock checking.
- Low-code and maintainable: no custom scripting, so the budget limit, approvers and options can be changed easily.
- Scalable and reusable: the flow can be packaged as an update set and extended to other standard requests.

Disadvantages

- Works only for standard laptop configurations; non-standard or special requests still need manual handling.
- Depends on accurate Inventory data; stale stock counts can give wrong stock-check results.
- Requires a ServiceNow instance and users with the correct roles, which may involve licensing cost in a real deployment.
- Email notifications depend on the instance's email configuration being set up correctly.
- Does not yet integrate with vendor purchasing or asset-management systems.

9. CONCLUSION

This project replaced the manual, follow-up-heavy handling of standard laptop requests with an automated workflow on the ServiceNow platform. The Laptop Order catalog item guides the employee to a valid configuration, Flow Designer applies the budget rule to auto-approve routine orders and route only exceptions to the Department Head, and the flow creates the fulfillment task for IT Support after checking stock. Stage-wise notifications and a reporting dashboard keep requesters and managers informed throughout.

The solution addresses the pain points identified in the ideation phase: requesters no longer wait without information, approvers are not distracted by routine approvals, and IT staff are freed from repetitive steps. Because it is built entirely with native low-code features, it is easy to maintain and can serve as a template for automating other standard IT requests.

10. FUTURE SCOPE

- Extend the same pattern to other standard requests such as monitors, mobile phones, peripherals and software licenses.
- Integrate with vendor or procurement systems to place purchase orders automatically when stock is low.
- Add automatic re-order alerts when the Inventory stock count falls below a threshold.
- Introduce role-based laptop presets so that new joiners get a pre-selected configuration for their job role.
- Add a Virtual Agent or Service Portal widget so employees can order and track laptops through chat.
- Add SLA tracking and escalation for fulfillment tasks that stay open too long.
- Package the catalog item and flow as an update set so other departments or client organizations can reuse it.

11. APPENDIX

Source Code

The solution uses no custom source code. It is configured with the native ServiceNow Service Catalog (catalog item, variables, Catalog UI Policy), Flow Designer (flow, actions and conditions), Notifications and Dashboards. The configuration can be exported as a ServiceNow update set (XML) and attached in the GitHub repository.

Dataset Link

No external dataset is used. The Inventory table holds sample laptop models with stock counts, created manually in the ServiceNow instance.

GitHub & Project Demo Link

GitHub Repository	https://github.com/Pradeepa-K-R/Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-with-Flow-Designer
Project Demo Video	https://drive.google.com/file/d/1GNUxUVFKwFWT2VjsTo1ggksaMkkrTZVr/view
ServiceNow Instance (PDI)	https://dev421839.service-now.com/login.do?user_name=admin&sys_action=sysverb_login&user_password=knDBw%3Dv%25B08R